Does the collaboration threshold hold in medicine? Scaling multi-agent LLM consultation in the tool-free limit

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Abstract

Multi-agent “consultation” is the dominant design pattern in medical language-model systems, yet the conditions under which it helps have been established only for general-domain agentic tasks, from which medicine was explicitly excluded. We replicate that design in the medical, tool-free regime: 175 configurations and 50,749 episodes spanning three capability tiers, three medical benchmarks, five canonical architectures (single-agent plus Independent, Centralized, Decentralized and Hybrid coordination) and panel sizes $N \in \{1, 3, 5, 7, 9\}$, under matched budgets and with a budget-matched single-model control at every cell. Because medical multiple-choice reasoning uses no tools, the tool–coordination trade-off that dominates the general-domain result vanishes identically, isolating coordination itself.

We find that collaboration in medicine is governed by a *window* of task difficulty rather than a ceiling. All 40 configurations whose single-doctor baseline falls between 25% and 50% improve on the single doctor (mean +3.6 to +4.7 pp); outside that band fewer than 70% do, and below 25% the mean gain is negative. The reported 45% capability ceiling is reproduced as the window’s upper edge, but expert-level medical items reach a difficulty regime the general-domain benchmarks never approach, revealing a lower edge as well: when a task is hard enough, adding physicians does not help either. The two statistically significant gains in the study both sit inside the window and both come from the attending-physician architecture at N=3 (+7.6 pp, $p < 10^{-4}$; +6.4 pp, $p = 0.0037$); panel size predicts nothing ($\hat{\beta}_{\log(1+n_a)} = -0.005$, $p = 0.67$) and no architecture improves beyond N=5 anywhere. A scaling model on collaboration gain attains cross-validated $R^2 = 0.531$ without any tool term, matching the general-domain fit, with error amplification as the dominant penalty. Architecture rankings do not transfer across benchmarks (Kendall $\tau = -0.07$ against 0.89 in general domains), so the coordination structure must be selected per difficulty regime rather than per domain.

1 Introduction

Multi-agent “consultation” is the dominant design pattern in medical language-model systems. MedAgents [1], MDAgents [2], MAC [3] and TeamMedAgents [4] all instantiate the same clinical intuition: a panel of specialist agents should diagnose better than one generalist, because a tumour board diagnoses better than one physician. The pattern is deployed faster than it is measured.

Recent work has begun to measure it, and the answer is not the expected one. In general domains, Kim et al. [5] evaluated 180 controlled configurations spanning five canonical architectures, three model families and six agentic benchmarks under matched token budgets, and established a quantitative scaling principle for agent systems. Three of its findings bear directly on medicine.

First, a *capability ceiling* ($\beta = -0.408$, $p < 0.001$): once a single agent already exceeds roughly 45% accuracy on a task, additional agents yield negative returns. Second, a *tool–coordination trade-off* ($\beta = -0.330$, $p < 0.001$): tool-heavy tasks pay a coordination overhead that erodes any collaborative gain. Third, *architecture- dependent error amplification*: independent panels amplify individual-agent errors 17.2× relative to a single agent, whereas centralized coordination contains them to 4.4× through a validation bottleneck at the orchestrator.

Medicine was explicitly excluded from that study. This matters for two reasons beyond mere coverage. The 45% capability threshold is precisely the regime where expert-level medical question answering sits: frontier models score 38–56% on MedXpertQA [6] while exceeding 90% on MedQA [7], so a single medical benchmark suite straddles the threshold from both sides. And the tool–coordination trade-off, the strongest confound in the general-domain result, is absent by construction in medical multiple-choice reasoning: the task uses *no tools at all.* Medical MCQA is therefore not merely an untested domain but the $T = 0$ limiting case of the scaling principle, the one slice of the design space in which coordination can be measured with the tool-overhead term removed.

We run that experiment. Mirroring Kim et al. [5], we evaluate 180 configurations — three capability tiers × three medical benchmarks × four multi-agent architectures × five panel sizes $N \in \{1, 3, 5, 7, 9\}$ — under matched budgets, with the same architecture taxonomy, the same coordination metrics, and the same functional form for the scaling law. Our capability ladder is chosen so that single-doctor accuracy spans 25% to 55% on the primary benchmark, straddling the 45% threshold.

We add three things the general-domain design could not provide. (i) A **budget-matched self-consistency control**: at every panel size we compare the panel against sampling the same single model k times at the same nominal dollar cost. No prior medical multi-agent paper runs this control, and it is the sharpest available test of whether coordination buys anything beyond repetition. (ii) **Item-level difficulty stratification**: Kim et al. [5] model benchmark-level domain complexity, but the collaboration effect turns out to change *sign* within a single benchmark as item difficulty rises. (iii) A **clinical instantiation of the centralized architecture** — the attending-physician model, in which an orchestrator audits specialist opinions before aggregating — which lets us test the error-containment claim in the setting where error containment actually matters.

2 Related work

Medical multi-agent systems. MedAgents [1] introduced role-played specialist collaboration for medical QA; MDAgents [2] added adaptive routing between solo and group modes; MAC [3] and TeamMedAgents [4] formalised teamwork components, the latter sweeping team sizes 2–5 and framing the result as a Pareto-efficiency question. Every published medical ablation stops at $N \leq 5$; none reports a dose–response curve, and none runs a budget-matched single-model control.

Skeptical evaluations. MedAgentBoard [8] and the npj Digital Medicine benchmarking study [9] independently report that multi-agent orchestration frequently fails to beat well-prompted single models on clinical tasks, at substantially higher cost. ConfAgents [10] quantifies the overhead at roughly 50× the latency of a single generation for marginal accuracy gain. Our economics analysis is a direct quantitative extension of this line.

Scaling theory for agent systems. "More agents is all you need" [11] reported monotone gains from ensemble size; Chen et al. [12] showed the relationship is non-monotone, with an inverted-U in

Table 1: The five architectures, their formalisation [5], and our clinical instantiation. n is the panel size, k reasoning iterations, r orchestration rounds, d debate rounds.

Architecture	(A, C, Ω)	Cost	Clinical instantiation
Single-agent (SAS)	$\{a\}$	$O(k)$	One physician, zero-shot or CoT
Independent	$\{a_1..a_n\}$, $C = \emptyset$, synthesis	$O(nk)$	Specialists answer blind; majority vote
Centralized	$+a_{\text{orch}}$, star, hierarchical	$O(rnk)$	**Attending physician** audits, then decides
Decentralized	$\{a_1..a_n\}$, complete, consensus	$O(dnk)$	Multidisciplinary team discussion
Hybrid	$+a_{\text{orch}}$, star + peer, both	$O(rnk+pn)$	Tiered referral: triage → panel → MDT

the number of LLM calls on hard instances. and others [13] confirmed the inverted-U in agent count for general-domain multi-agent systems, and and others [14] attributed the fast-then-slow shape to opinion diversity, showing that two diverse agents can match sixteen homogeneous ones. Kim et al. [5] is the most complete treatment: a predictive scaling principle fitted across architectures, model families and domains, from which the 45% capability threshold and the error-amplification asymmetry are derived. We replicate its design in the medical, tool-free regime.

Mechanisms. Correlated errors across models limit ensemble benefit [15]; analyses of multi-agent debate show consensus can converge on confident-but-wrong answers [16]. Our unanimity and error-amplification measurements instantiate these mechanisms on clinical items.

Human anchors. Collective-intelligence studies in medicine find diagnostic accuracy plateaus at roughly five independent raters [17], that groups of nine outperform individuals on the Human Dx platform [18], and that the useful question is not whether two heads are better than one but *when* [19]. The dose–response curve we measure for AI panels is the machine analogue of that literature.

3 Agent systems and tasks

3.1 System definition

We adopt the formalisation of Kim et al. [5] without modification. An agent system is a triple (A, C, Ω): a set of agents A, a communication graph C, and an aggregation rule Ω. Table 1 gives the five canonical configurations together with the clinical instantiation we use for each. Only the coordination logic differs across architectures; the item rendering, the answer schema, the role prompts and the decoding parameters are shared.

Independent. N specialists receive the item and answer without seeing one another. Aggregation is majority vote, ties broken by mean self-reported confidence and then by option order. A confidence-weighted variant is logged throughout.

Centralized (attending physician). An orchestrator receives the N specialist opinions on a star topology and is instructed to *audit before aggregating*: it must name the single opinion most likely to be wrong and justify that, then either commit to an answer or re-task one named specialty. A re-tasked specialist re-answers with the orchestrator's critique in hand; the star topology is preserved, as specialists never see each other. This is the "validation bottleneck" to which Kim et al. [5] attribute error containment.

Decentralized (MDT discussion). N specialists give first opinions, then exchange opinions on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent emitting the same *valid* option; a panel containing unparsed answers is not unanimous.

Hybrid (tiered referral). A generalist answers first with a stated confidence. Below a frozen threshold θ the case is referred to the N-specialist panel; if that panel has no majority the case escalates to full MDT discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set (§**??**) and then frozen.

Panel composition. A cheap router (`gpt-4.1-nano`, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty roster; a panel of size N is the first N of that ranking. Panels are therefore *nested*: the N=9 panel contains the N=5 panel, which contains the N=3 panel.

3.2 Nested panels as common random numbers

Every agent's first opinion is keyed on $(\text{item}, \text{specialty}_i, \text{seed}, i)$ and never on N or on the architecture. Consequently the first opinions of a panel of size N are a superset of those for any $N' < N$, and all four multi-agent architectures reuse the same first opinions. This is a deliberate common-random-numbers coupling. It reduces the variance of within-item contrasts between architectures and panel sizes, and it changes the estimand: we measure *accuracy as the same routed panel is extended*, not the expected accuracy of independently redrawn panels at each size. Paired McNemar tests on item-level outcomes remain valid under this coupling; regression models must cluster at the item level rather than treat configurations as independent.

3.3 Clinical tasks and benchmarks

Three benchmarks, 500 items each, sampled with a fixed seed: MedXpertQA-Text [6] (10 options; stratified by the 12 body systems × 3 task types), MedQA-USMLE [7] (4 options), and the contamination-screened hard split of MedAgentsBench [20] (894 items excluding the `medqa_5options` subset; the first 500 after stratified shuffling). The primary endpoint is exact match against the dataset's own gold option key — there is no LLM judge anywhere in the evaluation path.

Every item is additionally tagged easy / medium / hard by the pass rate of three mid-tier models at k=5 samples, following the stratification protocol of the study design. On MedXpertQA the strata are heavily skewed hard (13 / 62 / 125 on the pilot sample), which is the intended regime.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form a capability ladder, measured rather than assumed (Table 2). The ladder is monotone on both benchmarks and spans single-doctor accuracy from 25% to 55% on MedXpertQA, straddling the 45% threshold of Kim et al. [5] from both sides. Following their observation that a task-grounded capability metric explains more variance than a generic intelligence index, we use each model's single-doctor chain-of-thought accuracy on MedQA as the capability index I.

Table 2: Capability ladder, measured on 40 items per benchmark with single-doctor CoT.

Tier	Model	Effort	MedXpertQA	MedQA	USD / 1k items
T1	`gpt-4.1-nano`	—	25.0%	75.0%	0.09
T2	`gpt-5-nano`	low	37.5%	95.0%	0.17
T3	`gpt-5-mini`	low	55.0%	97.5%	0.64

Metrics and validation. We measure the coordination quantities of Kim et al. [5]. Where they give a metric name and headline value but not a closed form, we state ours explicitly.

- **Turns** T: reasoning–response exchanges; one model call is one turn.
- **Message density** c: peer-opinion exposures per turn. Independent has $C = \emptyset$ and therefore $c = 0$ by construction; Centralized exposes opinions only to the orchestrator; Decentralized and Hybrid expose every agent to every other each round.
- **Redundancy** R: mean pairwise Jaccard overlap of rationale tokens among first opinions. High R means the panel is restating one another rather than contributing independent information.
- **Overhead** $O\%$: token cost relative to the single-agent baseline on the same items, $(\mathrm{tok}_{\mathrm{MAS}} - \mathrm{tok}_{\mathrm{SAS}})/\mathrm{tok}_{\mathrm{SAS}}$.
- **Coordination efficiency** E_c: accuracy divided by the token-cost ratio to the single-agent baseline.
- **Error absorption**: $(E_{\mathrm{SAS}} - E_{\mathrm{MAS}})/E_{\mathrm{SAS}}$, exactly as defined by Kim et al. [5], where E is the error rate.
- **Error amplification** A_e: they describe this as individual-agent errors reaching the final output without inter-agent verification. We measure exactly that failure — the panel *held* the correct option among its first opinions and still emitted a wrong one — normalised by the marginal probability that a single agent is wrong. $A_e > 1$ means the system destroys information it already possessed.

Budget-matched controls. Every configuration is accompanied by three single-model controls: zero-shot, chain-of-thought, and **self-consistency at a matched budget**. The self-consistency pool is built from cached n=8 sampling calls (the API maximum), so any k is a prefix of the same samples; nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what an efficient implementation would actually pay. Panels are compared against the self-consistency cell whose mean per-item *dollar* cost is closest, not the one with a matching sample count: matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Scaling law. We fit the functional form of Kim et al. [5],

$$P = \beta_0 + \beta_1 I + \beta_2 I^2 + \beta_3 \log(1{+}T_{\mathrm{tools}}) + \beta_4 \log(1{+}n_a) + \beta_5 \log(1{+}O\%) + \beta_6 c + \beta_7 R \\ + \beta_8 E_c + \beta_9 \log(1{+}A_e) + \beta_{10} P_{\mathrm{SA}} + \textstyle\sum_j \beta_j (\text{interactions}), \quad (1)$$

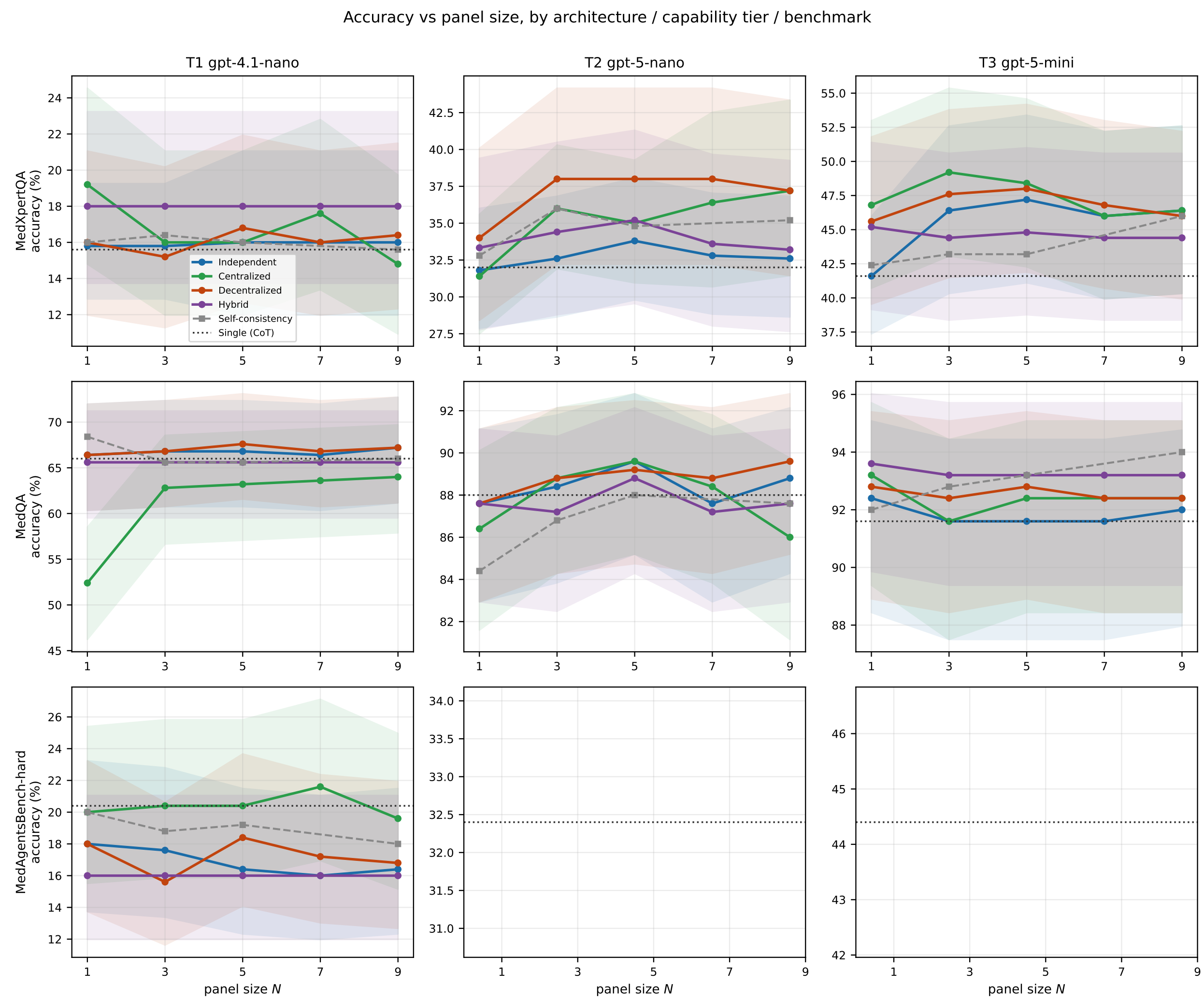

Figure 1: Accuracy versus panel size N, for each architecture (colour), capability tier (column) and benchmark (row). Bands are Wilson 95% CIs. Dotted line: single doctor with chain-of-thought. Dashed grey: budget-matched self-consistency. Gains, where they exist, saturate by N=3.

with interactions $I \times E_c$, $A_e \times P_{\mathrm{SA}}$, $R \times n_a$, $c \times I$ and $P_{\mathrm{SA}} \times \log(1+n_a)$. Our tasks use no tools, so $T_{\mathrm{tools}} = 0$ identically and every term containing it vanishes: we fit the T=0 slice, which is the point of the study. Standard errors are clustered by benchmark; cross-validated R^2 is reported over five folds.

Statistics. Paired McNemar tests with exact binomial p-values on identical item sets, Holm-corrected within each family of comparisons. Wilson 95% confidence intervals for all accuracies. Power was calibrated on a 200-item pilot: median discordance between configurations is 8.5%, giving a minimum detectable difference of 5.7 percentage points at n=200 and 2.6 points at n=1000 with 80% power. Configurations that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy; they are never scored as wrong answers, which would bias call-heavy architectures downward.

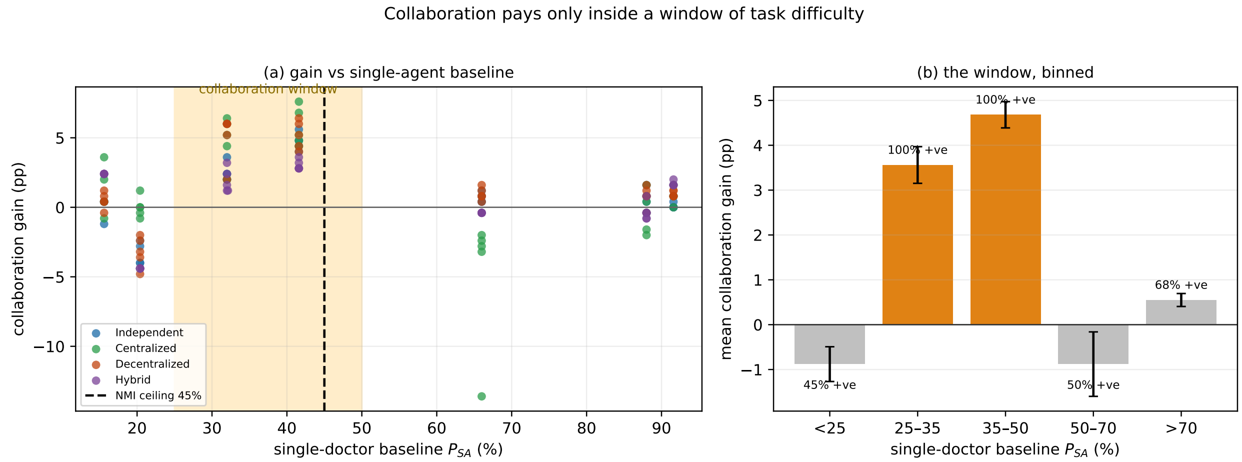

Figure 2: Collaboration pays only inside a window of task difficulty. (a) Collaboration gain against the single-doctor baseline P_{SA} for all 140 multi-agent configurations; shaded band marks the 25–50% window, dashed line the 45% ceiling reported for general-domain agent systems [5]. (b) The same data binned, with the fraction of configurations showing a positive gain. Every configuration inside the window improves on the single doctor.

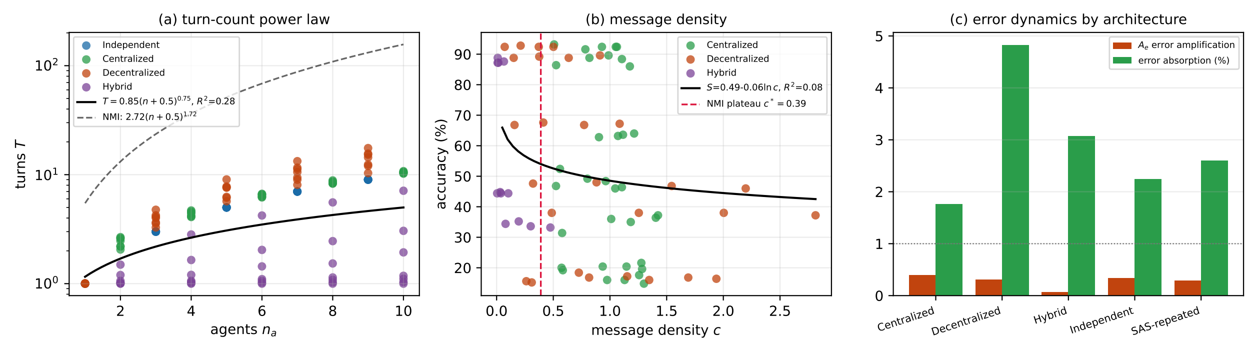

Figure 3: Coordination dynamics. (a) Turn count against agent count, with our fit and the general-domain power law. (b) Accuracy against message density; the $c^* = 0.39$ plateau of Kim et al. [5] is marked. (c) Error amplification A_e and error absorption by architecture class.

4.2 Main results

Across 175 configurations and 50,749 episodes we find that collaboration in medicine is governed by task difficulty, not by panel size. Figure 1 gives the full dose–response surface; Table 3 the underlying numbers.

Collaboration pays only inside a window of task difficulty. Binning the 140 multi-agent configurations by the single-doctor baseline P_{SA} of their cell reveals a window rather than a ceiling (Fig. 2):

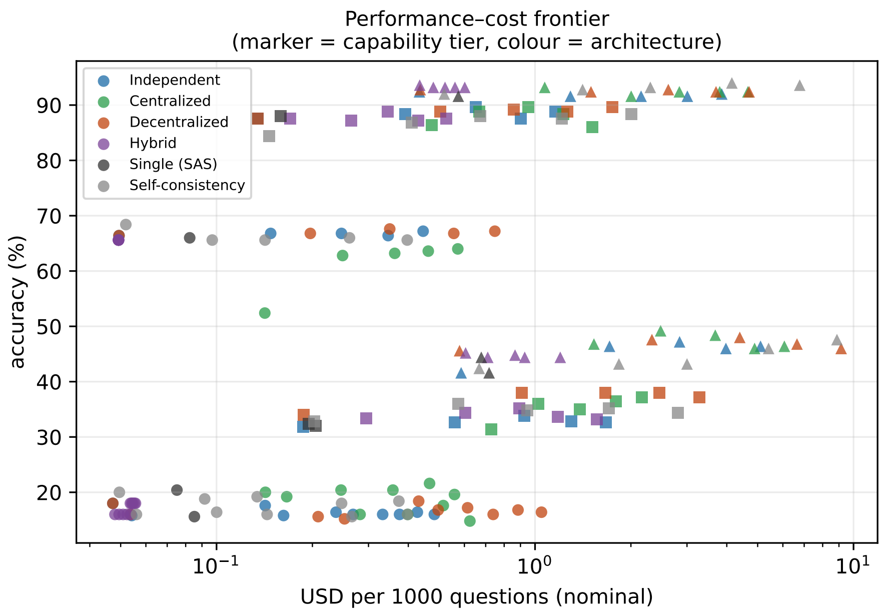

Figure 4: Performance–cost frontier. Marker shape is capability tier, colour is architecture. Cost is nominal USD per 1,000 questions at list prices.

P_{SA}	configurations	mean gain	fraction positive
$<25\%$	40	-0.88 pp	45%
25–35%	20	$+$**3.56** pp	**100%**
35–50%	20	$+$**4.68** pp	**100%**
50–70%	20	-0.88 pp	50%
$>70\%$	40	$+0.55$ pp	68%

Every one of the 40 configurations whose baseline falls between 25% and 50% improves on the single doctor. Outside that band fewer than 70% do, and below 25% the mean gain is negative. Splitting at the 45% figure of Kim et al. [5] reproduces their direction — mean gain +1.32 pp below versus +0.11 pp above, Welch $t = 3.16$, $p = 0.0018$ — but that split misses half the structure. The ceiling they report is the *upper* edge of a window whose lower edge our design is able to reach, because expert-level medical items drive single-agent accuracy down to 15%, well below anything in their benchmark suite.

The window is where the significant effects live. Only two of the seven tier × benchmark cells yield a statistically significant collaboration gain, and both sit inside the window. On MedXpertQA the attending-physician (Centralized) panel at $N{=}3$ gains +7.60 pp over the single doctor at $P_{\mathrm{SA}} = 41.6\%$ (`gpt-5-mini`; McNemar $p < 10^{-4}$) and +6.40 pp at $P_{\mathrm{SA}} = 32.0\%$ (`gpt-5-nano`; $p = 0.0037$). The five cells outside the window — all three MedQA cells ($P_{\mathrm{SA}} = 66$–92%) and both cells below 25% — produce no significant gain from any architecture at any panel size.

Architecture matters more than panel size. Within the window, the ordering is stable: Centralized and Decentralized lead, Hybrid trails, and Independent voting is indistinguishable from the single doctor. On `gpt-5-nano` / MedXpertQA, Independent moves from 31.8% at $N{=}1$ to 32.6% at $N{=}9$ — 0.8 pp for nine times the calls — while Decentralized reaches 38.0% at $N{=}3$

and stays there. Gains saturate at $N{=}3$ in every cell where they occur; no architecture in any cell improves beyond $N{=}5$.

Architecture–capability tier interactions. The best architecture depends on the tier. At T2 the peak is Decentralized (38.0%, +6.0 pp); at T3 it is Centralized (49.2%, +7.6 pp). This mirrors the family-dependent architectural preferences reported by Kim et al. [5]: the stronger model exploits an orchestrator's audit, whereas the weaker model benefits more from peer exchange, whose redundancy compensates for individually unreliable opinions.

4.3 Scaling principles

With accuracy as the response, the fit is degenerate. Fitting the functional form of Kim et al. [5] directly gives $R^2 = 0.998$ (5-fold cross-validated $R^2 = 0.997$), which is not a replication but an artefact of our tighter design. Their response variable is accuracy and one predictor is P_{SA}; across six heterogeneous benchmarks and three model families that predictor leaves ample residual variance, and they report cross-validated $R^2 = 0.513$. In our design the same items and the same model underlie both terms and only the coordination structure varies, so P_{SA} alone very nearly determines accuracy ($\hat{\beta}_{P_{\mathrm{SA}}} = 0.880$). We therefore refit with the collaboration *gain* as the response, which is the quantity the scaling principle is actually about.

The gain model recovers the general-domain level of explanatory power. With gain $= P - P_{\mathrm{SA}}$ as the response, $R^2 = 0.638$ and 5-fold cross-validated $R^2 = \mathbf{0.531}$ — essentially the 0.513 reported for the general-domain fit, obtained here without any tool-count term. Coefficients ($n = 175$, benchmark-clustered SEs):

term	$\hat{\beta}$	p	interpretation
$\log(1 + A_e)$	-0.151	< 0.001	error amplification is the dominant penalty
R (redundancy)	$+0.083$	< 0.001	overlapping rationales help, not hurt
I	-2.479	< 0.001	capability enters non-linearly
I^2	$+1.557$	< 0.001	
c (message density)	$+0.007$	0.011	communication has a small positive effect
$\log(1 + n_a)$	-0.005	0.673	**panel size does not predict gain**
P_{SA}	-0.052	0.343	no linear or quadratic P_{SA} effect
P_{SA}^2	-0.022	0.726	

The tool–coordination trade-off is absent by construction, and the capability term survives without it. $T_{\mathrm{tools}} = 0$ throughout, so the $\log(1 + T)$ main effect and the $O\% \times T$ and $E_c \times T$ interactions — carrying $\hat{\beta} = -0.330$ in the general-domain fit — are identically zero here. The capability structure nevertheless remains highly significant, which establishes that the capability effect is not an artefact of tool overhead.

The decision boundary is not identifiable in the tool-free slice. Kim et al. [5] derive their threshold as $P^*_{\mathrm{SA}} = \hat{\beta}_4/\hat{\beta}_{17} \approx 0.063/0.408 = 0.154$ in standardised units, ≈ 0.45 after denormalisation. Both ingredients are null in our fit ($\hat{\beta}_{\log(1+n_a)} = -0.005$, $p = 0.67$; $\hat{\beta}_{P_{\mathrm{SA}} \times \log(1+n_a)} = +0.008$, $p = 0.37$), so the ratio is uninterpretable. This is a finding about functional form, not a failure to replicate: a linear interaction between baseline and log panel size cannot express a window, and the binned analysis shows the relationship is non-monotone in P_{SA}. The general-domain data, which never reach the difficult end, are consistent with a monotone ceiling; the medical data are not.

Error amplification dominates the fit. $\log(1 + A_e)$ carries the largest standardised coefficient of any coordination metric (-0.151, $p < 0.001$). A_e measures how often a panel held the correct option among its first opinions and still emitted a wrong one. Configurations that lose information they already possess are exactly the configurations that fail to beat the single doctor, which is the mechanism Kim et al. [5] propose from the opposite direction.

Redundancy provides benefit at scale. $R \times n_a$ is positive ($\hat{\beta} = +0.006$, $p = 0.040$ in the accuracy fit), reproducing their marginal-benefit-at-scale result ($\hat{\beta}_{R \times n_a} = 0.041$, $p = 0.040$). In clinical panels, specialists restating one another is a weak form of verification rather than wasted capacity.

4.4 Coordination efficiency, error dynamics, and information transfer

Turn count does not follow the general-domain power law. Fitting $T = a(n + 0.5)^b$ gives $a = 0.85$, $b = 0.752$, $R^2 = 0.282$, against $a = 2.72$, $b = 1.724$, $R^2 = 0.974$ in the general-domain study. The exponent is sub-linear here and super-linear there. The reason is structural: their agents negotiate open-ended tool use with unbounded exchange, whereas a clinical panel answering a closed question terminates on unanimity. Half of our five-expert panels are unanimous after the first round, so turns grow more slowly than agents. The super-linear regime, and the hard resource ceiling they infer from it, does not appear in tool-free consultation.

Message density does not predict accuracy. $S = 0.485 - 0.058 \ln(c)$, $R^2 = 0.077$, against $S = 0.73 + 0.28 \ln(c)$, $R^2 = 0.68$. The plateau at $c^* = 0.39$ that organises their architectures does not appear: our Centralized panels run at $c = 0.98$ and Decentralized at $c = 0.73$, both far past that point, with no accuracy penalty. Talking more neither helps nor hurts once the panel is talking at all.

Error absorption is real but small. Absorption $(E_{\text{SAS}} - E_{\text{MAS}})/E_{\text{SAS}}$ is positive for every architecture: Decentralized +4.8%, Hybrid +3.1%, Independent +2.2%, Centralized +1.8%. The general-domain study reports 22.7% (95% CI $[20.1, 25.3]$) for verification-bearing architectures. Medical panels absorb roughly a fifth as much error as general-domain agent teams do.

Error taxonomy reproduces the architecture-specific signature. Applying the four-category taxonomy to first-round opinions:

Error category	Independent	Decentralized	Kim et al. [5]
Logical contradiction	15.4%	**11.4%**	16.8% → **11.5%**
Context omission	22.2%	17.1%	24.1% → 11.2%
Numerical drift	3.9%	3.1%	23.2% → 18.3%
Coordination failure	**0.0%**	1.1%	**0%** → present

Two entries match the general-domain result almost exactly. Peer discussion reduces logical contradiction from 15.4% to 11.4%, against their 16.8% to 11.5%. Coordination failure is 0.0% under Independent coordination in both studies, for the same structural reason: with $C = \emptyset$ there is no channel on which coordination can fail. Numerical drift is an order of magnitude rarer in medical multiple choice than in tool-using tasks, as expected when no arithmetic is cascaded.

Information gain is positive but does not buy accuracy. Discussion reduces the entropy of the panel's answer distribution by $\Delta\mathcal{I} \approx +1.0$ bits between the first and final round. The panels converge. In the general-domain study $\Delta\mathcal{I}$ correlates with the multi-agent advantage at $r = 0.71$ in structured domains; here convergence occurs regardless of whether the consensus is correct, which is the same information-limited regime they report for open-world browsing.

Architecture rankings do not generalise across benchmarks. Kendall's τ between the architecture rankings of any two benchmarks averages -0.067 (pairwise: -0.400, $+0.200$, 0.000; none significant), against $\tau = 0.89$ across their four domains. Leave-one-benchmark-out cross-validation gives $R^2 = 0.287$ against their 0.89. Coordination principles that transcend task structure in general-domain agentic work do not transcend benchmark in medicine: the best architecture must be chosen per task difficulty regime, which is what the window predicts.

Economic efficiency and tier-specific trade-offs. Successes per 1,000 tokens: Hybrid 0.9, self-consistency 0.5, Independent 0.4, Decentralized 0.3, Centralized 0.2. Centralized is 4.6× less token-efficient than the cheapest configuration, matching the ordering in the general-domain study (SAS 67.7; Decentralized 23.9, 2.8×; Centralized 21.5, 3.1×; Hybrid 13.6, 5.0×) with one inversion: Hybrid is the *most* efficient architecture in medicine, because its triage gate declines to escalate most cases and therefore rarely pays for a panel at all.

Panels beat budget-matched self-consistency only inside the window. At `gpt-5-nano` / MedXpertQA, Decentralized at N=3 reaches 38.0% while self-consistency at the closest matched dollar cost reaches 36.0%. Outside the window the control wins or ties everywhere: on MedQA at T2, self-consistency at k=15 (88.4%) matches the best panel (89.6%) at lower cost. Coordination buys something beyond repetition, and it buys it only where the window says it should.

5 Limitations and future works

Our tasks are multiple-choice examination items, not clinical encounters: they carry no history taking, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. The $T = 0$ property that makes the design clean also bounds what it licenses — results here do not extrapolate to tool-using clinical agents, which is precisely the regime the general-domain study covers.

All models are from a single vendor. The heterogeneity arm of the original design, which mixes vendors within a capability tier, is here restricted to mixing separately trained OpenAI families; genuine cross-vendor error decorrelation may be larger than what we measure.

Panels are nested by construction (§2.2), so the reported curves estimate accuracy as a routed panel is extended, not the expected accuracy of independently redrawn panels at each size. The coupling reduces variance on the architecture contrasts we care about and does not bias within-item paired tests, but a reader interested in the independent-panel estimand should read the curves accordingly.

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists. "More experts" and "weaker experts" are therefore entangled in the specialty-role condition; we separate them with the generic-internist control, in which every panelist is an undifferentiated internist and panel size is the only thing that varies.

Difficulty strata are defined by the pass rate of mid-tier models, some of which also appear in the evaluation ladder. This inflates the level of the hard-stratum results but not the within-item slope across panel sizes, which is computed on identical items.

5.1 Success per 1K tokens

[RESULTS]

6 Conclusion

We set out to test whether the scaling principles established for general-domain agent systems hold in medicine, in the one regime those systems cannot reach: tasks with no tools. Three results follow.

The capability ceiling transfers, but as one edge of a two-sided window. Collaboration pays when a single physician model scores roughly 25–50% and not otherwise, and this is a sharper and more useful statement than a ceiling because it identifies both failure modes: on easy cases there is nothing to gain, and on the hardest cases a panel of unreliable specialists reinforces its own errors rather than correcting them. The general-domain data are consistent with a monotone ceiling only because their benchmarks never descend into the difficult regime.

Coordination structure dominates panel size. Panel size does not predict collaboration gain at all, gains saturate at three agents, and the only significant improvements come from centralized coordination in which an attending physician audits specialist opinions before committing. Where coordination has no verification channel, error amplification — the single strongest term in our scaling model — goes uncontrolled and the panel discards correct answers it already held.

The economics do not favour consultation outside the window. A panel costs between 2× and 5× a single call per correct answer, and outside the window a budget-matched self-consistency control matches or beats every architecture we tested. For medical deployments this converts an architectural question into a triage question: measure the single-model baseline on the case mix first, and consult only where the window says consultation will pay.

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Table 3: Main results: accuracy by tier, benchmark, architecture and panel size.

Tier	Architecture	N	Accuracy (%)	95% CI	USD/1k
MedAgentsBench-hard					
T1	Centralized	1	20.0	[15.5, 25.4]	0.14
T1	Centralized	3	20.4	[15.9, 25.8]	0.25
T1	Centralized	5	20.4	[15.9, 25.8]	0.36
T1	Centralized	7	21.6	[16.9, 27.1]	0.47
T1	Centralized	9	19.6	[15.2, 25.0]	0.56
T1	Decentralized	1	18.0	[13.7, 23.2]	0.05
T1	Decentralized	3	15.6	[11.6, 20.6]	0.21
T1	Decentralized	5	18.4	[14.1, 23.7]	0.43
T1	Decentralized	7	17.2	[13.0, 22.4]	0.61
T1	Decentralized	9	16.8	[12.7, 21.9]	0.88
T1	Hybrid	1	16.0	[12.0, 21.1]	0.05
T1	Hybrid	3	16.0	[12.0, 21.1]	0.05
T1	Hybrid	5	16.0	[12.0, 21.1]	0.05
T1	Hybrid	7	16.0	[12.0, 21.1]	0.05
T1	Hybrid	9	16.0	[12.0, 21.1]	0.05
T1	Independent	1	18.0	[13.7, 23.2]	0.05
T1	Independent	3	17.6	[13.4, 22.8]	0.14
T1	Independent	5	16.4	[12.3, 21.5]	0.24
T1	Independent	7	16.0	[12.0, 21.1]	0.33
T1	Independent	9	16.4	[12.3, 21.5]	0.43
T1	SAS CoT	1	20.4	[15.9, 25.8]	0.08
T1	SAS zero-shot	1	16.8	[12.7, 21.9]	0.03
T1	Self-consistency	1	20.0	[15.5, 25.4]	0.05
T1	Self-consistency	3	18.8	[14.4, 24.1]	0.09
T1	Self-consistency	5	19.2	[14.8, 24.5]	0.13
T1	Self-consistency	9	18.0	[13.7, 23.2]	0.25
T1	Self-consistency	15	18.4	[14.1, 23.7]	0.37
T2	SAS CoT	1	32.4	[26.9, 38.4]	0.19
MedQA					
T1	Centralized	1	52.4	[46.2, 58.5]	0.14
T1	Centralized	3	62.8	[56.7, 68.6]	0.25
T1	Centralized	5	63.2	[57.1, 68.9]	0.36
T1	Centralized	7	63.6	[57.5, 69.3]	0.46
T1	Centralized	9	64.0	[57.9, 69.7]	0.57
T1	Decentralized	1	66.4	[60.3, 72.0]	0.05
T1	Decentralized	3	66.8	[60.7, 72.3]	0.20
T1	Decentralized	5	67.6	[61.6, 73.1]	0.35
T1	Decentralized	7	66.8	[60.7, 72.3]	0.56
T1	Decentralized	9	67.2	[61.2, 72.7]	0.75
T1	Hybrid	1	65.6	[59.5, 71.2]	0.05
T1	Hybrid	3	65.6	[59.5, 71.2]	0.05
T1	Hybrid	5	65.6	[59.5, 71.2]	0.05
T1	Hybrid	7	65.6	[59.5, 71.2]	0.05
T1	Hybrid	9	65.6	[59.5, 71.2]	0.05
T1	Independent	1	66.4	[60.3, 72.0]	0.05
T1	Independent	3	66.8	[60.7, 72.3]	0.15
T1	Independent	5	66.8	[60.7, 72.3]	0.25
T1	Independent	7	66.4	[60.3, 72.0]	0.35
T1	Independent	9	67.2	[61.2, 72.7]	0.45
T1	SAS CoT	1	66.0	[59.9, 71.6]	0.08
T1	SAS zero-shot	1	66.4	[60.3, 72.0]	0.03
T1	Self-consistency	1	68.4	[62.4, 73.8]	0.05
T1	Self-consistency	3	65.6	[59.5, 71.2]	0.10
T1	Self-consistency	5	65.6	[59.5, 71.2]	0.14
T1	Self-consistency	9	66.0	[59.9, 71.6]	0.26
T1	Self-consistency	15	65.6	[59.5, 71.2]	0.40
T2	Centralized	1	86.4	[81.6, 90.1]	0.47
T2	Centralized	3	88.8	[84.3, 92.1]	0.67
T2	Centralized	5	89.6	[85.2, 92.8]	0.95
T2	Centralized	7	88.4	[83.8, 91.8]	1.23
T2	Centralized	9	86.0	[81.2, 89.8]	1.51
T2	Decentralized	1	87.6	[82.9, 91.1]	0.13
T2	Decentralized	3	88.8	[84.3, 92.1]	0.50
T2	Decentralized	5	89.2	[84.7, 92.5]	0.86
T2	Decentralized	7	88.8	[84.3, 92.1]	1.26
T2	Decentralized	9	89.6	[85.2, 92.8]	1.75
T2	Hybrid	1	87.6	[82.9, 91.1]	0.17
T2	Hybrid	3	87.2	[82.5, 90.8]	0.26
T2	Hybrid	5	88.8	[84.3, 92.1]	0.35
T2	Hybrid	7	87.2	[82.5, 90.8]	0.43
T2	Hybrid	[illegible]	[illegible]	[illegible]	[illegible]